

МИНОБРНАУКИ РОССИИ
САНКТ-ПЕТЕРБУРГСКИЙ ГОСУДАРСТВЕННЫЙ
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ОТЧЕТ
по лабораторной работе №2
по дисциплине «Технологии хранения данных»
Тема: Kafka

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2026

Цель работы

Создать docker-compose проект для демонстрации работы apache kafka.

Описание решения

Приложение симулирует поток «гонок» и демонстрирует работу распределённой системы: продюсер генерирует события гонок и публикует их в Kafka, два консьюмера читают и обрабатывают эти события в реальном времени.

Архитектура:

- Producer (FastAPI): router.py — фоновый бесконечный цикл генерирует события для трёх гонок и публикует в соответствующие Kafka topics.
- Kafka: брокер + Zookeeper, управляется через docker-compose.yml.
- Consumers: consumer1.py — выводит live-события с эמודзи; consumer2.py — отслеживает лидеров и печатает сводку.
- Мониторинг: Kafdrop (веб UI) для просмотра топиков и сообщений.

Поток данных:

- Producer формирует JSON-сообщение и публикует в topic (f1_events, nascar_events, lemans_events).
- Консьюмеры подписаны на эти темы и обрабатывают сообщения (logging, подсчёт лидеров).

Формат сообщения (пример)

```
{  
  "race": "F1",  
  "driver": "Hamilton",  
  "position": 1,  
  "lap": 50,  
  "timestamp": "2026-05-20T..."  
}
```

Скриншоты

1. Запуск и состояние контейнеров

```
aleksandrusanov@Aleksandra-MacBook-Air: kafka % docker compose up -d --build
[+] Building 10.6s (27/27) FINISHED
=> [internal] load local bake definitions
=> reading from stdin 1.45s
=> [app internal] load build definition from Dockerfile
=> transferring dockerfile: 281B
=> [consumer1 internal] load build definition from Dockerfile.consumer1
=> transferring dockerfile: 279B
=> [consumer1 internal] load build definition from Dockerfile.consumer1
=> [app internal] load metadata for docker.io/library/python:3.11-slim
=> [app internal] load .dockerignore
=> transferring context: 2B
=> [app 1/5] FROM docker.io/library/python:3.11-slim@sha256:2c285c669cc837ea3b0cf1af23ea1932b7b5214f9c9d3aad22417446ad91cb4fb
=> resolve docker.io/library/python:3.11-slim@sha256:2c285c669cc837ea3b0cf1af23ea1932b7b5214f9c9d3aad22417446ad91cb4fb
=> [consumer2 internal] load build context
=> transferring context: 2.04kB
=> [app internal] load build context
=> transferring context: 13.56kB
=> [consumer1 internal] load build context
=> transferring context: 1.47kB
=> CACHED [consumer2 2/5] WORKDIR /consumer
=> CACHED [consumer1 3/5] COPY requirements.txt ./
=> CACHED [consumer2 4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [consumer1 5/5] COPY consumer/consumer1.py .
=> CACHED [app 2/5] WORKDIR /app
=> CACHED [app 3/5] COPY requirements.txt ./
=> CACHED [app 4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [app 5/5] COPY app
=> CACHED [consumer2 3/5] COPY requirements.txt ./
=> CACHED [consumer2 4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [consumer1 5/5] COPY consumer/consumer2.py .
=> [consumer1] exporting to image
=> exporting layers
=> exporting manifest sha256:148717614e1ead8bd343f72719783bb08998f6392e3f388b12ee346973ccdd6
=> exporting config sha256:47732191d140b0c7f5aa69278f34df556f61122793b5ef3f65a99d873f9e6a3e
=> exporting attestation manifest sha256:2a658ea7228532dafd285a6540af3c4c1a3bd1815c45752a9db974dad4f695e
=> exporting manifest list sha256:2aac04626c769782cef614c0b410b98b6e88bef789cc638eadd778db814974c7
=> naming to docker.io/library/kafka-consumer1:latest
=> unpacking to docker.io/library/kafka-consumer1:latest
=> [app] exporting to image
=> exporting layers
=> exporting manifest sha256:13fca181f370bea5e723f462ade6f1c4932a96883c8fc5897d9cfd43e29f7bb
=> exporting config sha256:036f6d07a7a1f1f63619a79e70b11589c6f5a26c62b0f08e5098c1df49919
=> exporting attestation manifest sha256:1f6a9a8550939222f5a638ff91b5b680a6c9cc9fedeb8a81c67692298
=> exporting manifest list sha256:5b4579db950b9aebd9e3ac8720b9e8c4a85b3a12af08e66828132a104307a
=> naming to docker.io/library/kafka-app:latest
=> unpacking to docker.io/library/kafka-app:latest
=> [consumer2] exporting to image
=> exporting layers
=> exporting manifest sha256:329444dd71ff2483a81998781557bd57f43f3b9cf8ac397fbdb0e8b18cc4e8d
=> exporting config sha256:1dfe1671c8a8e8b7da7cc578f2b5ad77474731c83bd9f99ab232b74da91dea
=> exporting attestation manifest sha256:992724f18a86d13a53c462d0f81aa5d4fa7359176bc86a482ec9fc81286470
=> exporting manifest list sha256:18b32134a99cc39222445403187acd04534016ab4472a8a81aae8338923b5b7
=> naming to docker.io/library/kafka-consumer2:latest
=> unpacking to docker.io/library/kafka-consumer2:latest
=> [consumer1] resolving provenance for metadata file
=> [app] resolving provenance for metadata file
=> [consumer2] resolving provenance for metadata file
[+] Up 11/11
✔ Image kafka-consumer2 Built 10.6s
✔ Image kafka-app Built 10.6s
✔ Image kafka-consumer1 Built 10.6s
✔ Network kafka_default Created 0.0s
✔ Container zookeeper Started 0.5s
✔ Container kafka_healthy Healthy 14.2s
✔ Container kafka-init Exited 57.2s
✔ Container kafdrop Started 0.7s
✔ Container app Started 57.2s
✔ Container consumer1 Started 57.2s
✔ Container consumer2 Started 57.2s
aleksandrusanov@Aleksandra-MacBook-Air: kafka % docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
app                 kafka-app            "/bin/sh -c 'uvicorn..."  app        About a minute ago    Up 9 seconds    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
consumer1          kafka-consumer1      "/bin/sh -c 'python3..."  consumer1  About a minute ago    Up 9 seconds
consumer2          kafka-consumer2      "/bin/sh -c 'python3..."  consumer2  About a minute ago    Up 9 seconds
kafdrop            obsidiandynamics/kafdrop:3.31.0  "/kafdrop.sh"          kafdrop    About a minute ago    Up About a minute    0.0.0.0:9080->9080/tcp, [::]:9080->9080/tcp
kafka              gcr.io/itlms-public/mc-addons/kafka:2.8.0-debian-10-r65  "/opt/bitnami/script..." kafka      About a minute ago    Up About a minute (healthy)    0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp
zookeeper          gcr.io/itlms-public/mc-addons/zookeeper:3.7.0-debian-10-r70  "/opt/bitnami/script..." zookeeper  About a minute ago    Up About a minute    0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp
```

Результат docker compose up -d --build и вывод
docker compose ps

2. Просмотр логов consumer1

```
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Logano at P3 on lap 1 @ 2026-05-20T09:22:19.967532
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Larson at P4 on lap 1 @ 2026-05-20T09:22:19.982839
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Johnson at P1 on lap 1 @ 2026-05-20T09:22:19.947405
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Elliott at P2 on lap 1 @ 2026-05-20T09:22:19.963780
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Porsche Team at P1 on lap 1 @ 2026-05-20T09:22:19.987940
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Ferrari Team at P2 on lap 1 @ 2026-05-20T09:22:20.019460
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Toyota Team at P3 on lap 1 @ 2026-05-20T09:22:20.022385
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: BMW Team at P4 on lap 1 @ 2026-05-20T09:22:20.027360
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Hamilton at P1 on lap 1 @ 2026-05-20T09:22:19.835109
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Verstappen at P2 on lap 1 @ 2026-05-20T09:22:19.936103
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Sainz at P3 on lap 1 @ 2026-05-20T09:22:19.939835
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Leclerc at P4 on lap 1 @ 2026-05-20T09:22:19.944839
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Hamilton at P1 on lap 2 @ 2026-05-20T09:22:22.034334
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Verstappen at P2 on lap 2 @ 2026-05-20T09:22:22.055551
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Sainz at P3 on lap 2 @ 2026-05-20T09:22:22.080212
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Leclerc at P4 on lap 2 @ 2026-05-20T09:22:22.107799
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Johnson at P1 on lap 2 @ 2026-05-20T09:22:22.115470
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Elliott at P2 on lap 2 @ 2026-05-20T09:22:22.125280
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Logano at P3 on lap 2 @ 2026-05-20T09:22:22.134692
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Larson at P4 on lap 2 @ 2026-05-20T09:22:22.142394
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Porsche Team at P1 on lap 2 @ 2026-05-20T09:22:22.144164
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Toyota Team at P3 on lap 2 @ 2026-05-20T09:22:22.160138
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Ferrari Team at P2 on lap 2 @ 2026-05-20T09:22:22.156426
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: BMW Team at P4 on lap 2 @ 2026-05-20T09:22:22.162860
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Hamilton at P1 on lap 3 @ 2026-05-20T09:22:24.166571
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Verstappen at P2 on lap 3 @ 2026-05-20T09:22:24.175016
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Sainz at P3 on lap 3 @ 2026-05-20T09:22:24.193834
consumer1 | [CONSUMER1] | 🏁 F1 UPDATE: Leclerc at P4 on lap 3 @ 2026-05-20T09:22:24.207901
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Johnson at P1 on lap 3 @ 2026-05-20T09:22:24.219366
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Elliott at P2 on lap 3 @ 2026-05-20T09:22:24.288495
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Logano at P3 on lap 3 @ 2026-05-20T09:22:24.299124
consumer1 | [CONSUMER1] | 🏁 NASCAR UPDATE: Larson at P4 on lap 3 @ 2026-05-20T09:22:24.305791
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Porsche Team at P1 on lap 3 @ 2026-05-20T09:22:24.309604
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Ferrari Team at P2 on lap 3 @ 2026-05-20T09:22:24.310914
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: Toyota Team at P3 on lap 3 @ 2026-05-20T09:22:24.319821
consumer1 | [CONSUMER1] | 🏁 LeMans UPDATE: BMW Team at P4 on lap 3 @ 2026-05-20T09:22:24.329928
```

Поток логов consumer1 (docker compose logs -f consumer1)

3. Просмотр логов consumer2

```

consumer2 [CONSUMER2] 🏆 CURRENT LEADERS:
=====
consumer2 F1      → Hamilton      | Lap 1 | Messages: 4
consumer2 NASCAR → Johnson      | Lap 1 | Messages: 4
consumer2 LeMans → Porsche Team | Lap 1 | Messages: 4
consumer2
consumer2
consumer2 [CONSUMER2] 🏆 CURRENT LEADERS:
=====
consumer2 F1      → Hamilton      | Lap 2 | Messages: 8
consumer2 NASCAR → Johnson      | Lap 2 | Messages: 8
consumer2 LeMans → Porsche Team | Lap 2 | Messages: 8
consumer2
consumer2
consumer2 [CONSUMER2] 🏆 CURRENT LEADERS:
=====
consumer2 F1      → Hamilton      | Lap 3 | Messages: 12
consumer2 NASCAR → Johnson      | Lap 3 | Messages: 12
consumer2 LeMans → Porsche Team | Lap 3 | Messages: 12
consumer2
consumer2
consumer2 [CONSUMER2] 🏆 CURRENT LEADERS:
=====
consumer2 F1      → Hamilton      | Lap 4 | Messages: 16
consumer2 NASCAR → Johnson      | Lap 4 | Messages: 16
consumer2 LeMans → Porsche Team | Lap 4 | Messages: 16
consumer2
consumer2
consumer2 [CONSUMER2] 🏆 CURRENT LEADERS:
=====
consumer2 F1      → Hamilton      | Lap 5 | Messages: 20
consumer2 NASCAR → Johnson      | Lap 5 | Messages: 20
consumer2 LeMans → Porsche Team | Lap 5 | Messages: 20
consumer2
consumer2
consumer2 [CONSUMER2] 🏆 CURRENT LEADERS:
=====
consumer2 F1      → Hamilton      | Lap 6 | Messages: 24
consumer2 NASCAR → Elliott      | Lap 6 | Messages: 24
consumer2 LeMans → Toyota Team  | Lap 6 | Messages: 24
consumer2
consumer2
consumer2 [CONSUMER2] 🏆 CURRENT LEADERS:
=====
consumer2 F1      → Hamilton      | Lap 7 | Messages: 28
consumer2 NASCAR → Elliott      | Lap 7 | Messages: 28
consumer2 LeMans → Toyota Team  | Lap 7 | Messages: 28
consumer2

```

Поток логов consumer2 (docker compose logs -f consumer2)

4. Kafdrop


Kafdrop

3.31.0 [2023-03-20T17:07:14.026Z]

Kafka Cluster Overview

Bootstrap servers	kafka:9092
Total topics	4
Total partitions	56
Total preferred partition leader	100 %
Total under-replicated partitions	0

Brokers

ID	Host	Port	Rack	Controller	Number of partitions (% of total)
1001	kafka	9092	-	Yes	56 (100 %)

Topics [ACLs](#)

Name	Partitions	% Preferred	# Under-replicated	Custom Config
_consumer_offsets	50	100 %	0	Yes
f1_events	2	100 %	0	No
lemans_events	2	100 %	0	No
nascar_events	2	100 %	0	No

New

Kafdrop — overview

5. Kafdrop: fl_events

Kafdrop

Topic Messages: **fl_events**

First Offset: 0 Last Offset: 470 Size: 470

Partition: 0 Offset: 0 # messages: 100 Key format: DEFAULT Message format: DEFAULT View Messages

Offset: 0 Key: empty Timestamp: 2026-05-20 09:22:22.055 Headers: empty
{"race": "F1", "driver": "Verstappen", "position": 2, "lap": 2, "timestamp": "2026-05-20T09:22:22.055551"}

Offset: 1 Key: empty Timestamp: 2026-05-20 09:22:22.080 Headers: empty
{"race": "F1", "driver": "Sainz", "position": 3, "lap": 2, "timestamp": "2026-05-20T09:22:22.080212"}

Offset: 2 Key: empty Timestamp: 2026-05-20 09:22:22.107 Headers: empty
{"race": "F1", "driver": "Leclerc", "position": 4, "lap": 2, "timestamp": "2026-05-20T09:22:22.107799"}

Offset: 3 Key: empty Timestamp: 2026-05-20 09:22:24.167 Headers: empty
{"race": "F1", "driver": "Hamilton", "position": 1, "lap": 3, "timestamp": "2026-05-20T09:22:24.166571"}

Offset: 4 Key: empty Timestamp: 2026-05-20 09:22:26.334 Headers: empty
{"race": "F1", "driver": "Hamilton", "position": 1, "lap": 4, "timestamp": "2026-05-20T09:22:26.334388"}

Offset: 5 Key: empty Timestamp: 2026-05-20 09:22:26.339 Headers: empty
{"race": "F1", "driver": "Verstappen", "position": 2, "lap": 4, "timestamp": "2026-05-20T09:22:26.339418"}

Offset: 6 Key: empty Timestamp: 2026-05-20 09:22:30.515 Headers: empty
{"race": "F1", "driver": "Verstappen", "position": 2, "lap": 6, "timestamp": "2026-05-20T09:22:30.515591"}

Offset: 7 Key: empty Timestamp: 2026-05-20 09:22:30.520 Headers: empty
{"race": "F1", "driver": "Sainz", "position": 4, "lap": 6, "timestamp": "2026-05-20T09:22:30.520579"}

Offset: 8 Key: empty Timestamp: 2026-05-20 09:22:32.562 Headers: empty
{"race": "F1", "driver": "Hamilton", "position": 1, "lap": 7, "timestamp": "2026-05-20T09:22:32.562038"}

Offset: 9 Key: empty Timestamp: 2026-05-20 09:22:32.571 Headers: empty
{"race": "F1", "driver": "Verstappen", "position": 2, "lap": 7, "timestamp": "2026-05-20T09:22:32.571032"}

Сообщения в партии 0 topic fl_events

6. Kafdrop: lemans_events

Kafdrop

Topic Messages: **lemans_events**

First Offset: 0 Last Offset: 516 Size: 516

Partition: 0 Offset: 0 # messages: 100 Key format: DEFAULT Message format: DEFAULT View Messages

Offset: 0 Key: empty Timestamp: 2026-05-20 09:22:22.144 Headers: empty
{"race": "LeMans", "driver": "Porsche Team", "position": 1, "lap": 2, "timestamp": "2026-05-20T09:22:22.144164"}

Offset: 1 Key: empty Timestamp: 2026-05-20 09:22:22.160 Headers: empty
{"race": "LeMans", "driver": "Toyota Team", "position": 3, "lap": 2, "timestamp": "2026-05-20T09:22:22.160138"}

Offset: 2 Key: empty Timestamp: 2026-05-20 09:22:22.162 Headers: empty
{"race": "LeMans", "driver": "BMW Team", "position": 4, "lap": 2, "timestamp": "2026-05-20T09:22:22.162868"}

Offset: 3 Key: empty Timestamp: 2026-05-20 09:22:24.309 Headers: empty
{"race": "LeMans", "driver": "Porsche Team", "position": 1, "lap": 3, "timestamp": "2026-05-20T09:22:24.309684"}

Offset: 4 Key: empty Timestamp: 2026-05-20 09:22:26.378 Headers: empty
{"race": "LeMans", "driver": "Porsche Team", "position": 1, "lap": 4, "timestamp": "2026-05-20T09:22:26.378707"}

Offset: 5 Key: empty Timestamp: 2026-05-20 09:22:26.392 Headers: empty
{"race": "LeMans", "driver": "Ferrari Team", "position": 2, "lap": 4, "timestamp": "2026-05-20T09:22:26.392788"}

Offset: 6 Key: empty Timestamp: 2026-05-20 09:22:26.414 Headers: empty
{"race": "LeMans", "driver": "Toyota Team", "position": 3, "lap": 4, "timestamp": "2026-05-20T09:22:26.414392"}

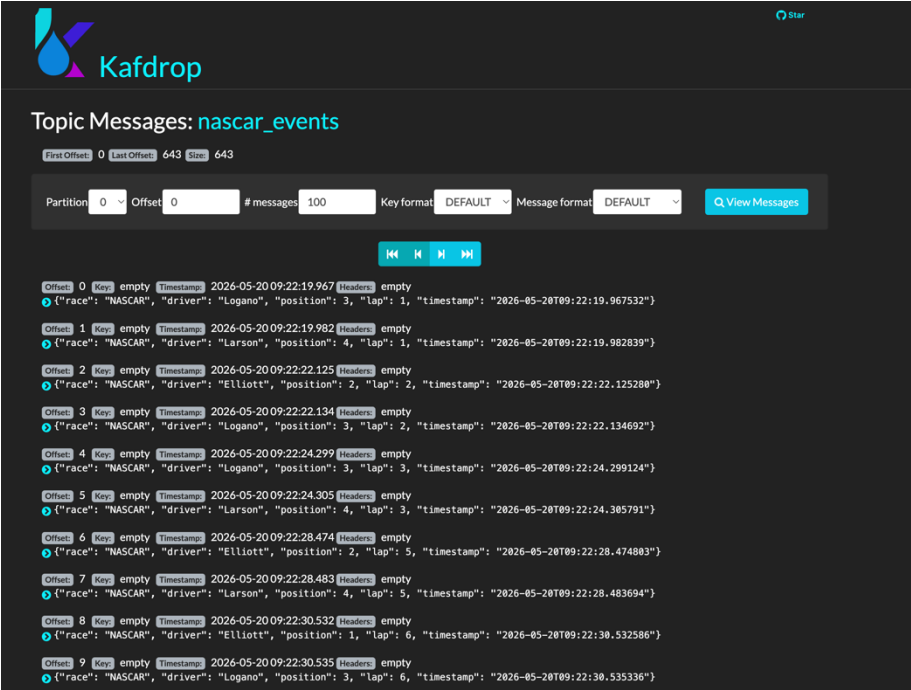
Offset: 7 Key: empty Timestamp: 2026-05-20 09:22:28.503 Headers: empty
{"race": "LeMans", "driver": "BMW Team", "position": 4, "lap": 5, "timestamp": "2026-05-20T09:22:28.503103"}

Offset: 8 Key: empty Timestamp: 2026-05-20 09:22:30.553 Headers: empty
{"race": "LeMans", "driver": "Ferrari Team", "position": 2, "lap": 6, "timestamp": "2026-05-20T09:22:30.553136"}

Offset: 9 Key: empty Timestamp: 2026-05-20 09:22:30.555 Headers: empty
{"race": "LeMans", "driver": "Toyota Team", "position": 1, "lap": 6, "timestamp": "2026-05-20T09:22:30.554974"}

Сообщения в партии 0 topic lemans_events

7. Kafdrop: nascar_events



Kafdrop

Topic Messages: **nascar_events**

First Offset: 0 Last Offset: 643 Size: 643

Partition: 0 Offset: 0 # messages: 100 Key format: DEFAULT Message format: DEFAULT [View Messages](#)

Navigation: ⏪ ⏴ ⏵ ⏩

Offset	Key	Timestamp	Headers	Message
0	empty	2026-05-20 09:22:19.967	empty	<pre>{ "race": "NASCAR", "driver": "Logano", "position": 3, "lap": 1, "timestamp": "2026-05-20T09:22:19.967532" }</pre>
1	empty	2026-05-20 09:22:19.982	empty	<pre>{ "race": "NASCAR", "driver": "Larson", "position": 4, "lap": 1, "timestamp": "2026-05-20T09:22:19.982839" }</pre>
2	empty	2026-05-20 09:22:22.125	empty	<pre>{ "race": "NASCAR", "driver": "Elliott", "position": 2, "lap": 2, "timestamp": "2026-05-20T09:22:22.125280" }</pre>
3	empty	2026-05-20 09:22:22.134	empty	<pre>{ "race": "NASCAR", "driver": "Logano", "position": 3, "lap": 2, "timestamp": "2026-05-20T09:22:22.134692" }</pre>
4	empty	2026-05-20 09:22:24.299	empty	<pre>{ "race": "NASCAR", "driver": "Logano", "position": 3, "lap": 3, "timestamp": "2026-05-20T09:22:24.299124" }</pre>
5	empty	2026-05-20 09:22:24.305	empty	<pre>{ "race": "NASCAR", "driver": "Larson", "position": 4, "lap": 3, "timestamp": "2026-05-20T09:22:24.305791" }</pre>
6	empty	2026-05-20 09:22:28.474	empty	<pre>{ "race": "NASCAR", "driver": "Elliott", "position": 2, "lap": 5, "timestamp": "2026-05-20T09:22:28.474803" }</pre>
7	empty	2026-05-20 09:22:28.483	empty	<pre>{ "race": "NASCAR", "driver": "Larson", "position": 4, "lap": 5, "timestamp": "2026-05-20T09:22:28.483694" }</pre>
8	empty	2026-05-20 09:22:30.532	empty	<pre>{ "race": "NASCAR", "driver": "Elliott", "position": 1, "lap": 6, "timestamp": "2026-05-20T09:22:30.532586" }</pre>
9	empty	2026-05-20 09:22:30.535	empty	<pre>{ "race": "NASCAR", "driver": "Logano", "position": 3, "lap": 6, "timestamp": "2026-05-20T09:22:30.535336" }</pre>

Сообщения в партии 0 topic nascar_events

Выводы

В ходе работы был создан и успешно запущен docker-compose проект для демонстрации работы Apache Kafka. Реализован producer на FastAPI, который генерирует и отправляет события о гоночных заездах в Kafka topics, а также два consumer-а, обрабатывающих эти сообщения в реальном времени. Корректность работы системы подтверждается тем, что сообщения поступают в Kafka, читаются consumer-ами и отображаются в Kafdrop, что позволяет визуально контролировать состояние топиков и поток данных.